

Vaibhav Kumar

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Work Experience

Software Intern | Security Metrix and Reporting | IBM

JUN 2024 – JULY 2024

- Developed and implemented automated security metrics dashboards to monitor vulnerabilities, compliance, and threat trends across multiple systems.
 - Integrated scanning tools like Trivy, Bandit, and OWASP ZAP into DefectDojo to generate detailed vulnerability reports and track remediation progress.
 - Conducted real-time analysis of security incidents, providing actionable insights to enhance system defenses and reduce potential attack surfaces.
 - Streamlined the reporting process by creating custom security reports, improving visibility and decision-making for security stakeholders.
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PROJECTS

Visual packet tracker | Visual packet tracker

JULY 2023

- Implemented packet capture and analysis features using protocols like TCP/IP and UDP to identify potential threats and performance bottlenecks.
- Optimized tool performance by integrating efficient data parsing algorithms, resulting in a 30% reduction in packet processing time.

Anomaly Detection | Anomaly Detection techniques

mar 2024

- Developed and implemented advanced anomaly detection algorithms for Intrusion Detection/Prevention Systems (IDS/IPS), significantly improving the identification of zero-day attacks and abnormal network behaviors.
- Enhanced the accuracy of threat detection by integrating machine learning models with IDS/IPS, reducing false positive rates by 20% and improving overall network security.

Sorting Visualizer | Sorting Algorithm

NOV 2023

- Developed an interactive sorting algorithm visualizer using HTML, CSS, and JavaScript, allowing users to visually explore and compare algorithms like Quick Sort, Merge Sort, and Bubble Sort in real-time.
 - Optimized the visualizer for performance and responsiveness, providing clear visual feedback on algorithm efficiency and time complexity, enhancing user understanding of sorting concepts.
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SKILLS

Programming Languages: C/C++ • Python

Cybersecurity Tools (Familiarity): Burp Suite • Nmap • Wireshark • Metasploit • EDR (MS DEFENDER)

Cybersecurity Concepts: Incident Response • Incident Handling • Email Security • Malware Analysis (static and Dynamic) • Threat Hunting

Penetration Testing: Test for common network and web application vulnerabilities.

CERTIFICATIONS

Prepare for the CompTIA Security+ (SY0-701) Certification

Achieved April 2024

Ethical Hacking Essentials (EHE) from EC-Council

Achieved June 2023

Network Defense Essentials (NDE) from EC-Council

Achieved Sep 2023

Fundamental of information security from Infosys

Achieved June 2024

Introduction to cybersecurity tools and cyber-attacks from Coursera

Achieved Aug 2024

EDUCATIONS

University of Petroleum and Energy Studies, Dehradun	2021-2025
Bachelor of Technology (B.Tech) in Computer Science C.G.P.A : 7.75	
R.L.S.C.S.V.M.I. College JASPUR, U.S.Nagar	2019-2020
12 th (UK Board) Percentage: 84%	
R.L.S.C.S.V.M.I. College JASPUR, U.S.Nagar	2017-2018
High School (UK Board) Percentage: 86%	

LEARNINGS

HACKERRANK/ LEETCODE

- Solved DSA easy and intermediate problems
- Improved skills in C++

TRYHACKME

Solve Offensive and Defensive Labs

- Learned about Cyber Security concepts like Reconnaissance, Exploitation, Scanning
- Use Nmap for network reconnaissance, uncovering hosts, open ports, and vulnerabilities.
- Use Wireshark for network traffic analysis, troubleshooting, and detecting anomalies.
- Utilized Burp Suite for web app security, intercepting requests and identifying flaws like SQLi and XSS

ACHIEVEMENT

- Participated Hack on Amazon - Season 4.
- Participated Focus on Code organized by the K. Ramakrishnan College of Engineering.
- Participated Capture the Flag organized by RV College of Engineering.
- Participated Anveshanam`24: Capture the Flag organized by Indian Institute of Technology (IIT), Jammu.
- Participated in NCIIPC-AICTE Pentathlon 2024.